

Basic Elements of Python

Character Set

1. **Alphabets** → A to Z, a to z, _ (underscore).
2. **Digits** → 0 to 9
3. **Blank Spaces** → Spacebar, Tab and Enter.
4. **Special Symbols** → +, -, *, /, %, @, #, <, >, ? etc.
5. **Binary Symbols** → E, §, ∧, μ etc.

Keywords/ Reserve Words

Keywords are the predefined words which represent some functionality.
There are 35 keywords in Python:

Keywords

No.	Keyword	No.	Keyword
1	False	19	in
2	None	20	is
3	True	21	lambda
4	and	22	nonlocal
5	as	23	not
6	assert	24	or
7	async	25	pass
8	await	26	raise
9	break	27	return
10	class	28	try
11	continue	29	while
12	def	30	with
13	del	31	yield
14	elif	32	match
15	else	33	case
16	except	34	finally
17	for	35	from
18	global		

To display list of all keywords

```
1  import keyword
2  print(keyword.kwlist)
3  print("Total keywords:", len(keyword.kwlist))
4
```

Import => keyword

Keyword => module (library)

Print() => function

Kwlist => Attribute

Variable

A variable is a name that is used to store a value in memory.

It acts like a container or label that holds data which can be used and changed during program execution.

```
1  name = "Rajat"  
2  age  = 20  
3
```

Identifier

Identifier is a name which is defined by the programmer such as variable name, function name, string name etc.

Rules of Identifier

1. Identifiers can contain only letters, digits, and underscores (_).
2. They cannot start with a digit.
3. They cannot be keyword.
4. Identifiers are case sensitive.
5. Blank spaces and special symbols are not allowed.

Comments

Comments are the ignorable part of our Python code that are ignored by the interpreter.

Comments are used to explain the code.

There are 2 types of comments in Python:

1. **Single Line comment:** Starts with # Symbol
2. **Multi Line comment:** We usually use triple quotes ('' or ''') for this purpose.